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Radiative transition probabilities, lifetimes, and dipole moments for all vibrational levels in the $X\,^1\Sigma^+$ state of ^{7}LiH[a)]

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The recently reported [J. Chem. Phys. **66**, 5412, **67**, 4785 (1977)] hybrid potential curve for the $X\,^1\Sigma^+$ state of LiH is combined with the *ab initio* dipole moment function of Docken and Hinze [J. Chem. Phys. **57**, 4936 (1972)] to calculate radiative transition probabilities between all ($v = 0$–23) vibrational levels of the $X\,^1\Sigma^+$ ground state of ^{7}LiH. In addition, the spontaneous emission lifetimes for all the vibrational–rotational levels with $J = 0$, 1, and 10 are given, as are the dipole moments for all vibrational levels ($J = 0$).

I. INTRODUCTION

The simplest neutral heteronuclear molecule, LiH, has been the object of intense theoretical and spectroscopic interest[1] since the 1930's. We have embarked on a long-term project to fully understand the spectroscopy of the three lowest singlet states ($X^1\Sigma^+$, $A^1\Sigma^+$, and $B^1\Pi$) of the LiH for a variety of reasons discussed elsewhere.[2,3]

As an initial step in this project, we have constructed a full potential curve (Fig. 1) for the $X^1\Sigma^+$ ground state of ^{7}LiH and from this curve calculated the vibrational (and vibrotational) energy levels of the isotopic lithium hydrides.[2] In this paper we summarize our calculations for the radiative transition probabilities among all vibrational levels of the $X^1\Sigma^+$ ($0 \le v'' \le 23$) state of ^{7}LiH and compare with available experimental data. In addition to interest in these quantities as such, we are also interested in two photon excitation of the $A^1\Sigma^+$ and $B^1\Pi$ states via near-resonant highly vibrationally excited states of $X^1\Sigma^+$ LiH and these results may be combined with our previous A–X and B–X results[2] to estimate these two photon probabilities. A more complete compilation of the results of these calculations is also available from one of the authors (W.C.S.). We also present radiative lifetimes for all vibrotational levels with $J = 0$, 1, and 10 and dipole moments for all vibrational levels ($J = 0$).

In the present manuscript, we first examine the appropriate formulas for absorption and emission transition probabilities and lifetimes for vibrational transitions in the $X^1\Sigma^+$ ground state of LiH in Sec. II. In Sec. III we discuss construction of the dipole moment function. Results of the transition probability, lifetime, and dipole moment calculations are summarized in Sec. IV. In Sec. V we discuss the physical interpretation of these results and compare with other (more limited) theoretical and experimental results.

II. RADIATIVE TRANSITION PROBABILITIES AND LIFETIMES

There are four basic kinds of spectroscopic intensity measurement, depending on whether absorption or emission is involved, and whether photon or energy flux is detected. The absolute intensities in each case involve a power of the transition frequency $\nu_{v',J';v'',J''}$

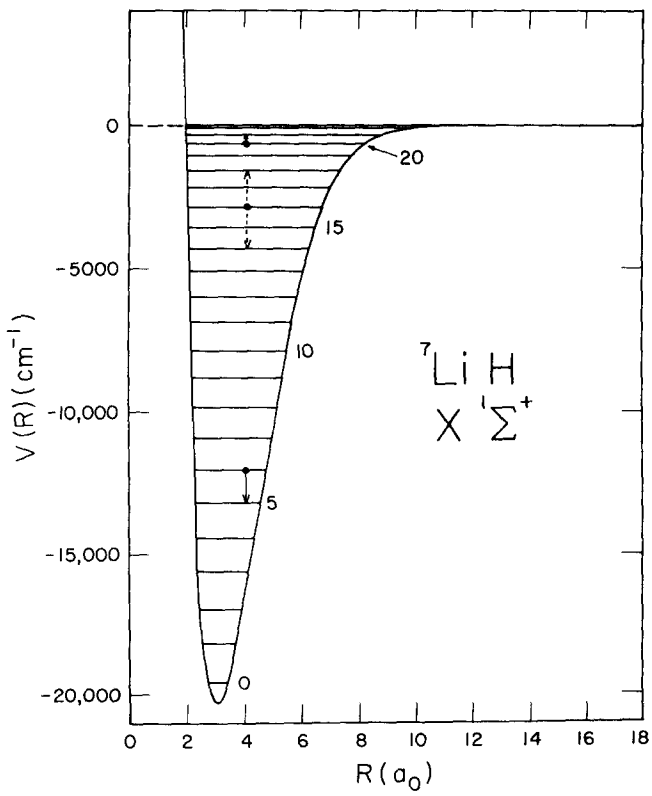


FIG. 1. The vibrational levels $v = 0$–23 of the Ref. 2 hybrid $X\,^1\Sigma^+$ potential energy curve of ^{7}LiH. Also shown in solid lines are the strongest photon absorption and emission fundamental ($|\Delta v| = 1$) transitions and in dashed lines the strongest photon absorption and emission first overtone ($|\Delta v| = 2$) transitions, as calculated in this work.

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TABLE I. The dipole moment function $\mu(R)$ in atomic units for the $X\,^1\Sigma^+$ ground state of ^{7}LiH.

Region	Range of R	Function $\mu(R)$	Reference
I	$0.0 \le R \le 15.0$	Tensioned spline fit	Table II
II	$15.0 \le R$	$18000\,R^{-7}$	a

[a]This value is the long range expansion from the separated atom limit value of zero; see text.

$= \hbar^{-1}[E'(v', J') - E''(v'', J'')]$ times the square of a dipole moment matrix element

$$\mu_{v',J';v'',J''} \equiv \int_0^\infty \psi_{v',J'}(R)\mu(R)\psi_{v'',J''}(R)\,dR$$

[where the upper and lower state wave functions $\psi_{v',J'}$ and $\psi_{v'',J''}$ are normalized and where $\mu(R)$ is the dipole moment function discussed below in Sec. III]. Note that if $v' = v''$ and $J' = J''$, we have simply the dipole moment expectation value $\langle\mu\rangle$ in the v', J' level. For the case of absorption measured by a decrease in photon flux, the transition probability is proportional to $|\mu_{v',J';v'',J''}|^2$ (no frequency dependence) and commonly expressed as line strength S or Einstein B coefficient. For absorption measured by a decrease in energy flux, the transition probability is proportional to $|\mu|^2\nu$; the absorption oscillator strength f or absorption coefficient α is commonly used in that case. For emission, the transition probabilities go as $|\mu|^2\nu^3$ and $|\mu|^2\nu^4$ for photon and energy detectors, respectively; the Einstein A coefficient is commonly used in the former case. An excellent discussion of the different quantities with conversion factors is given by Wiese, Smith, and Glennon.[4] A final quantity is the Franck–Condon factor

$$q_{v',J';v'',J''} = \left| \int_0^\infty \psi_{v',J'}(R)\,\psi_{v'',J''}(R)\,dR \right|^2$$

but these are not of much interest for the vibrational (as opposed to vibronic) transitions discussed here.

We shall restrict our attention here mainly to $J' = J'' - 1 = 0$ (although we report lifetimes for $J' = 0$, 1, and 10). We shall here report transition dipole moment matrix elements, $\mu_{v',0;v'',1}$ [in units of $(e\text{Å})$]; and Einstein A coefficients $[A_{v',0;v'',1}(\text{sec}^{-1})]$ (spontaneous photon emission intensities). The transition frequencies $\nu_{v',v''}$ are available using the energies listed in Ref. 2. The radiative lifetimes $\tau_{v',J'}$ are given for all bound v' with $J' = 0$, 1, and 10. The dipole moment expectation values $(\langle\mu\rangle \equiv \mu_{v,0;v,0})$ are given for all vibrational levels in atomic units.

III. THE DIPOLE MOMENT FUNCTION

The dipole moment function $\mu(R)$ has been constructed in a manner similar to that used previously for potential curves (e.g., in Ref. 2). The $\mu(R)$ function consists of two regions, given in Table I.

Region I is the region of predominant significance in our calculations. It consists of a cubic spline fit to the points given in Table II (see Fig. 2), which (except at 0.0, 1.0, 1.5, and 15.0 a_0) are the highly regarded *ab initio* results of Docken and Hinze.[5] The value of 0.0 a_0 is the united atom limit; at 1.0 and 1.5 a_0, from Brown and Shull,[6] and at 15 a_0, the long-range value (see below). The function and its first derivative are continuous everywhere, including at the Region II boundary. It is our opinion that the $\mu(R)$ in the physically important region from 2 to 12 a_0 is unlikely to be anywhere in error by as much as 20%. Finally it should be noted that the change in $\mu(R)$ with internuclear distance is quite sizeable.

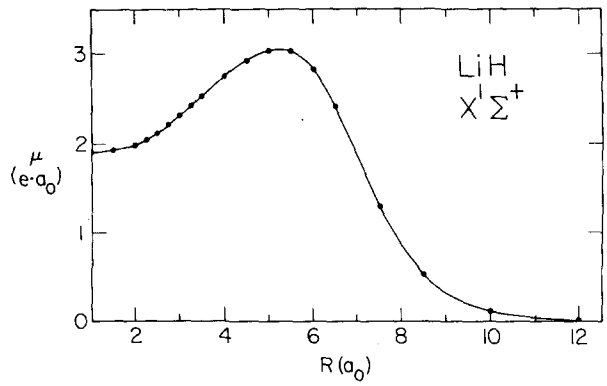


FIG. 2. Dipole moment function $\mu(R)$ (in atomic units) for the $X\,^1\Sigma^+$ ground state of ^{7}LiH, based primarily on the *ab initio* results of Docken and Hinze[6] (see text).

In the asymptotic Region II, we have used $\mu(R)$

TABLE II. The dipole moment function $\mu(R)$ (in a.u.) for the LiH $X\,^1\Sigma^+$ state in Region I (see Table I) consists of a tensioned cubic spline fit to the points listed below. Except for the points at $R = 0.0$ (the united atom limit), 1.0 and 1.5 a_0 (from Brown and Shull[6]) and $R = 15.0$ (the long-range limit) as discussed in the text, the points are from the work of Docken and Hinze.[5]

R (a_0)	$\mu(R)$ ($e\,a_0$)
0.0	0.0
1.0	1.9034
1.5	1.9185
2.0	1.9687
2.25	2.0340
2.5	2.1159
2.75	2.2092
3.0	2.3102
3.25	2.4153
3.5	2.5238
4.0	2.7353
4.5	2.9128
5.0	3.0236
5.5	3.0150
6.0	2.8182
6.5	2.4117
7.5	1.2911
8.5	0.5156
10.0	0.1130
12.0	0.0166
15.0	0.000105

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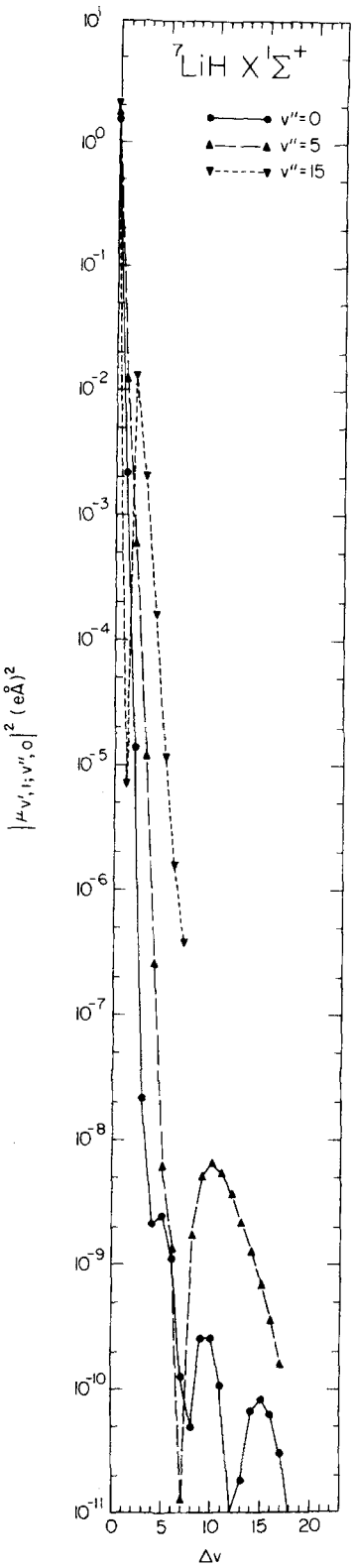


FIG. 3. Dipole moment matrix elements squared, $|\mu_{v',1;v'',0}|^2$, in $(e\text{ Å})^2$ units, for three selected lower levels, $v''=0$, 5, and 15, in the $X\,^1\Sigma^+$ state of ^{7}LiH. Note that the photon absorption rate is proportional to this quantity. The photon absorption is strongest for $\Delta v=0$, a pure rotational transition, and generally drops drastically as Δv increases. However, for higher v'' (e.g., 15), the fundamental ($\Delta v=1$) transition is not necessarily the strongest. Also, after decreasing roughly an order of magnitude per quantum from $\Delta v=1$ to 6, decreases for $\Delta v>6$ become much less rapid (with some oscillatory increases).

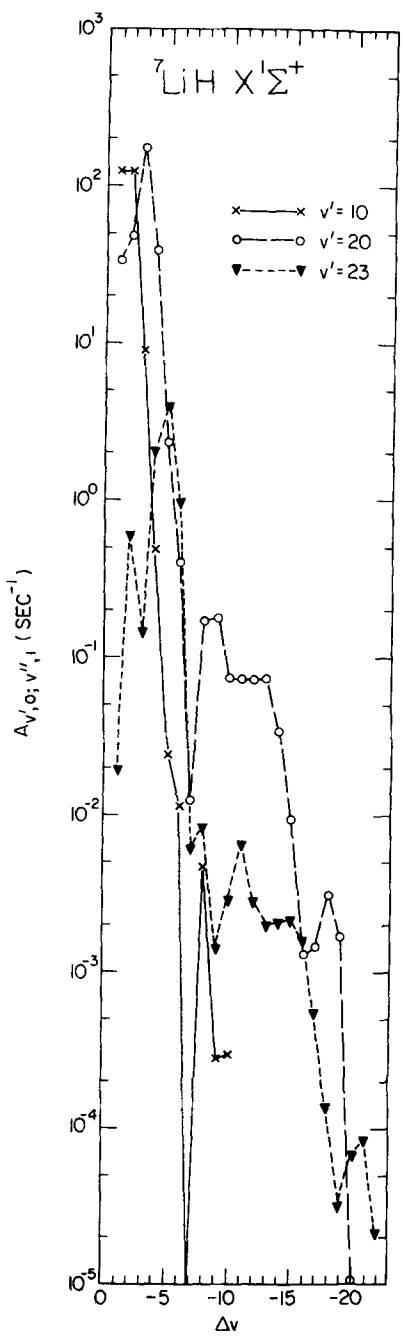


FIG. 4. Einstein $A_{v',0;v'',1}$ coefficients (spontaneous emission probabilities, in sec^{-1}) for three selected upper levels, $v'=10$, 20, and 23, in the $X\,^1\Sigma^+$ state of ^{7}LiH. Starting from $v'=2$ in Table III, one notes the first overtone emission ($\Delta v=-2$) becoming more and more competitive with fundamental emission ($\Delta v=-1$) until they are nearly equal for $v'=10$ as shown. For higher levels, even larger $|\Delta v|$ values are favored: the second overtone ($\Delta v=-3$) for $v'=20$ and the fourth (!) overtone ($\Delta v=-5$) for the long-range level $v'=23$. Note also the slower decrease in A value magnitude as Δv becomes larger, just as was found in Fig. 3. However, because of the ν^3 factor in A, the effect is more pronounced in emission.

$= 18\,000\,R^{-7}$. This D_7 value is our estimate based on the work of Byers–Brown and Whisnant.[7] Again, this region is not physically significant except for the highest level.

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TABLE III. Dipole moment absorption matrix elements $\mu_{v',J'=0;v'',J''=1}$ [in (eÅ) units, in upper right including pure rotational transitions on the diagonal] and Einstein A spontaneous emission coefficients $A_{v',J'=0;v'',J''=1}$ (in sec^{-1} units, in lower left) for the $X\,^1\Sigma^+$–$X\,^1\Sigma^+$ bands of ^{7}LiH. A coefficients are in exponential notation; e.g., 4.593(1) = 45.93.

	v'' ($J''=1$)							
v' ($J'=0$)	0	1	2	3	4	5	6	7
0	1.237290	0.046917	−0.003712	0.000147	0.000046	−0.000050	0.000034	−0.000011
1	4.593(1)	1.260081	0.067312	−0.006591	0.000430	−0.000058	−0.000002	0.000032
2	2.155(0)	8.288(1)	1.282077	0.082252	−0.010386	0.000798	−0.000100	0.000056
3	1.622(−2)	6.084(0)	1.149(2)	1.305193	0.094545	−0.014773	0.001496	−0.000167
4	1.104(−3)	1.082(−1)	1.367(1)	1.372(2)	1.328514	0.104430	−0.019315	0.002345
5	3.792(−3)	7.206(−3)	3.199(−1)	2.486(1)	1.532(2)	1.350390	0.112329	−0.024610
6	2.948(−3)	1.933(−4)	1.953(−2)	9.396(−1)	3.843(1)	1.614(2)	1.372119	0.118302
7	4.231(−4)	1.871(−3)	9.908(−3)	4.487(−2)	2.001(0)	5.597(1)	1.613(2)	1.392369
8	4.199(−4)	2.690(−3)	7.689(−4)	1.468(−2)	1.175(−1)	3.760(0)	7.600(1)	1.548(2)
9	2.433(−3)	4.469(−4)	2.203(−3)	6.710(−3)	2.218(−2)	2.500(−1)	6.237(0)	9.828(1)
10	3.064(−3)	2.826(−4)	4.823(−3)	6.371(−6)	1.138(−2)	2.457(−2)	4.953(−1)	9.173(0)
11	1.534(−3)	1.614(−3)	2.020(−3)	4.437(−3)	9.610(−4)	1.026(−2)	2.345(−2)	8.535(−1)
12	9.766(−5)	1.829(−3)	3.441(−5)	8.469(−3)	3.420(−3)	1.665(−3)	1.542(−3)	2.534(−2)
13	4.122(−4)	7.919(−4)	4.656(−4)	6.217(−3)	1.408(−2)	3.410(−3)	4.434(−7)	2.348(−3)
14	1.731(−3)	1.620(−5)	7.679(−4)	2.394(−3)	1.904(−2)	2.209(−2)	5.153(−3)	7.346(−3)
15	2.465(−3)	4.550(−4)	2.448(−4)	4.865(−4)	1.631(−2)	3.869(−3)	2.662(−2)	1.536(−2)
16	2.071(−3)	1.715(−3)	3.440(−5)	7.168(−5)	1.076(−2)	4.095(−2)	5.526(−2)	3.309(−2)
17	1.130(−3)	2.781(−3)	7.890(−4)	7.895(−5)	6.106(−3)	3.331(−2)	7.058(−2)	5.795(−2)
18	3.651(−4)	3.025(−3)	2.039(−3)	3.324(−4)	3.310(−3)	2.351(−2)	6.623(−2)	7.906(−2)
19	3.635(−5)	2.509(−3)	2.963(−3)	8.895(−4)	1.927(−3)	1.529(−2)	5.076(−2)	8.503(−2)
20	1.050(−5)	1.675(−3)	3.089(−3)	1.450(−3)	1.283(−3)	9.347(−3)	3.389(−2)	7.355(−2)
21	6.250(−5)	9.286(−4)	2.480(−3)	1.582(−3)	9.092(−4)	5.319(−3)	2.022(−2)	5.193(−2)
22	6.467(−5)	3.960(−4)	1.390(−3)	1.070(−3)	5.207(−4)	2.452(−3)	9.547(−3)	2.687(−2)
23	4.731(−6)	2.220(−5)	8.594(−5)	7.055(−5)	3.302(−5)	1.422(−4)	5.577(−4)	1.615(−3)

	8	9	10	11	12	13	14	15
0	−0.000007	0.000016	−0.000016	0.000010	−0.000002	−0.000004	0.000008	−0.000009
1	−0.000029	0.000011	0.000005	−0.000012	0.000011	−0.000007	0.000001	0.000004
2	−0.000006	−0.000030	0.000034	−0.000019	0.000004	0.000005	−0.000007	0.000004
3	0.000066	−0.000036	−0.000008	0.000036	−0.000040	0.000030	−0.000017	0.000007
4	−0.000313	0.000081	−0.000047	0.000003	0.000037	−0.000055	0.000054	−0.000044
5	0.003445	−0.000512	0.000079	−0.000037	0.000004	0.000042	−0.000073	0.000082
6	−0.030377	0.004750	−0.000802	0.000068	0.000018	−0.000028	0.000056	−0.000087
7	0.122171	−0.036657	0.006160	−0.001155	0.000069	0.000101	−0.000096	0.000095
8	1.410342	0.123727	−0.043585	0.007681	−0.001579	0.000103	0.000197	−0.000184
9	1.424(2)	1.425234	0.122517	−0.051443	0.009391	−0.002076	0.000223	0.000289
10	1.227(2)	1.249(2)	1.436476	0.117789	−0.060465	0.011453	−0.002600	0.000479
11	1.243(1)	1.498(2)	1.031(2)	1.443036	0.108589	−0.070779	0.014254	−0.003141
12	1.336(0)	1.608(1)	1.798(2)	7.832(1)	1.443716	0.093803	−0.082224	0.018323
13	3.735(−2)	1.943(0)	2.053(1)	2.122(2)	5.282(1)	1.436985	0.072215	−0.094335
14	1.918(−2)	8.467(−2)	2.567(0)	2.697(1)	2.440(2)	2.933(1)	1.420865	0.042371
15	3.484(−2)	4.439(−2)	2.264(−1)	3.133(0)	3.726(1)	2.699(2)	1.100(1)	1.392638
16	3.939(−2)	9.246(−2)	3.501(−2)	4.853(−1)	3.821(0)	5.402(1)	2.814(2)	1.058(0)
17	5.164(−2)	7.678(−2)	1.689(−1)	2.207(−4)	7.662(−1)	5.223(0)	8.037(1)	2.673(2)
18	6.569(−2)	7.082(−2)	1.511(−1)	1.621(−1)	6.313(−2)	9.907(−1)	8.720(0)	1.174(2)
19	7.418(−2)	7.398(−2)	1.025(−1)	2.378(−1)	4.088(−2)	2.479(−1)	1.249(0)	1.788(1)
20	7.241(−2)	7.180(−2)	7.466(−2)	1.788(−1)	1.699(−1)	1.231(−2)	4.010(−1)	2.307(0)
21	5.885(−2)	5.837(−2)	5.600(−2)	1.054(−1)	1.844(−1)	1.605(−2)	1.403(−1)	4.995(−1)
22	3.403(−2)	3.383(−2)	3.258(−2)	4.922(−2)	1.097(−1)	3.517(−2)	3.427(−2)	1.609(−1)
23	2.135(−3)	2.126(−3)	2.065(−3)	2.879(−3)	6.804(−3)	2.905(−3)	1.376(−3)	8.587(−3)

	16	17	18	19	20	21	22
0	0.000008	−0.000006	0.000003	−0.000001	0.000000	0.000001	−0.000001
1	−0.000008	0.000010	−0.000010	0.000008	−0.000007	0.000005	−0.000003
2	0.000001	−0.000006	0.000009	−0.000010	0.000010	−0.000009	0.000006
3	−0.000003	0.000002	−0.000004	0.000006	−0.000008	0.000008	−0.000006
4	0.000033	−0.000023	0.000015	−0.000011	0.000008	−0.000007	0.000005
5	−0.000075	0.000061	−0.000047	0.000036	−0.000026	0.000019	−0.000013
6	0.000105	−0.000105	0.000093	−0.000075	0.000058	−0.000043	0.000028
7	−0.000106	0.000118	−0.000121	0.000115	−0.000099	0.000079	−0.000055
8	0.000154	−0.000144	0.000138	−0.000131	0.000118	−0.000100	0.000073
9	−0.000300	0.000227	−0.000185	0.000164	−0.000145	0.000121	−0.000088
10	0.000294	−0.000426	0.000339	−0.000246	0.000188	−0.000149	0.000107
11	0.000833	0.000128	−0.000465	0.000464	−0.000358	0.000254	−0.000165
12	−0.003832	0.001195	−0.000193	−0.000306	0.000454	−0.000417	0.000300

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TABLE III (*Continued*

	v'' ($J''=1$)						
v' ($J'=0$)	16	17	18	19	20	21	22
13	0.024341	−0.005031	0.001547	−0.000567	0.000035	0.000210	−0.000239
14	−0.106122	0.033161	−0.007446	0.002028	−0.000885	0.000427	−0.000173
15	0.002662	−0.115536	0.045529	−0.012459	0.003387	−0.001251	0.000610
16	1.348595	−0.048335	−0.118890	0.061239	−0.022069	0.007689	−0.002959
17	1.748(0)	1.283899	−0.111078	−0.109550	0.077339	−0.037716	−0.017191
18	2.187(2)	1.231(1)	1.192821	−0.183493	−0.076553	0.082412	−0.050360
19	1.569(2)	1.370(2)	2.678(1)	1.067326	−0.258048	−0.008180	0.048744
20	3.890(1)	1.711(2)	4.738(1)	3.367(1)	0.895563	−0.313696	0.083278
21	7.296(0)	7.159(1)	1.175(2)	1.337(0)	2.381(1)	0.667810	−0.298400
22	1.435(0)	2.057(1)	6.846(1)	2.154(1)	6.458(0)	6.252(0)	0.371701
23	6.107(−3)	9.635(−1)	3.974(0)	2.039(0)	1.430(−1)	5.931(−1)	1.931(−2)

IV. RADIATIVE TRANSITION PROBABILITY AND LIFETIME RESULTS

The dipole moment function matrix elements and Einstein A coefficients (spontaneous photon emission intensities in sec^{-1}) as defined in Sec. II are given in Table III for all vibrational levels v' and v'' of the $X^1\Sigma^+$ state of ^{7}LiH. Selected examples of the dipole moment matrix elements and Einstein A coefficients are also displayed in Figs. 3 and 4.

These calculations have been carried out using an extensively modified version[8] of the diatomic intensity factor program of Zare and co-workers,[9] based on the Numerov–Cooley numerical solution of the radial Schrödinger equation which has previously been extensively checked.[2] Except for transitions involving the higher levels, the calculations are typically carried out with 0.004 Å grid from 0.5 to 8.5 Å, and have been extensively checked for convergence with respect to grid spacing and inner and outer limits. For levels whose wave functions are well described by this grid (i.e., $v \leq 21$), we feel all figures in Table III are numerically significant except for A coefficients of magnitude 10^{-5} or less.

For the higher levels, a wider range of internuclear distance must be used because of the slow decay with increasing distance of the wave functions near dissociation in the classically forbidden region.[3] We show in Fig. 5 an example of the convergence for the energy of the highest $J=0$ level, $v=23$, along with three Franck–Condon factors involving this level. The grid spacing remained fixed in this case as the outermost grid position R_{max} was varied from 14.5 to 22.5 Å. It is clear that integration out to large R_{max} is needed for conver-

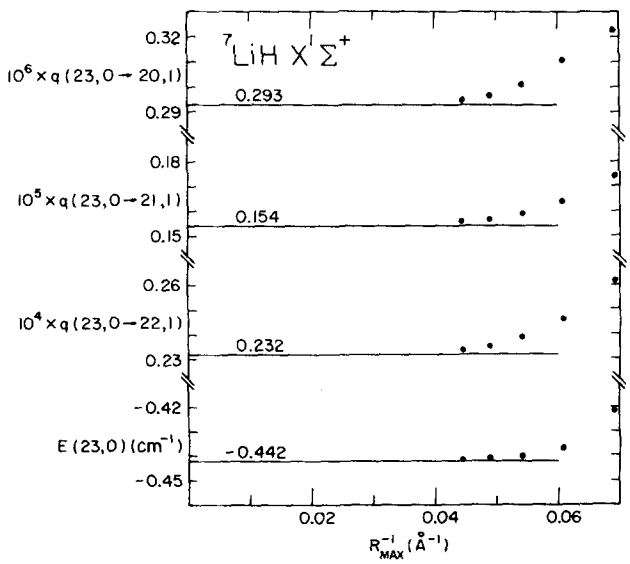


FIG. 5. Demonstration of the convergence of the energy, $E_{23,0}$ (in cm^{-1}), of the $v=23$, $J=0$ long-range level and of Franck–Condon factors, $q_{23,0;v'',1}$, for $v''=20$, 21, and 22, as a function of $1/R_{max}$, where R_{max} is the outermost point (in Å) of the numerical grid used solution of the radial Schrödinger equation.

TABLE IV. Spontaneous radiative lifetimes (in milliseconds) calculated for various v', J' levels of the $X\,^1\Sigma^+$ states of ^{7}LiH.

v' \ $J''=0$	1	10
0	∞	83710	61.91
1	21.77	21.81	17.56
2	11.76	11.78	10.88
3	8.264	8.279	8.147
4	6.623	6.635	6.727
5	5.605	5.616	5.803
6	4.980	4.990	5.230
7	4.559	4.568	4.823
8	4.261	4.269	4.526
9	4.045	4.052	4.297
10	3.886	3.894	4.113
11	3.756	3.763	3.948
12	3.629	3.633	3.771
13	3.478	3.481	3.570
14	3.301	3.302	3.348
15	3.108	3.110	3.127
16	2.932	2.933	2.943
17	2.810	2.803	2.840
18	2.787	2.788	2.871
19	2.935	2.938	3.127
20	3.396	3.404	3.848
21	4.490	4.507	5.932
22	7.984	8.073	...
23	128.7	...	...

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TABLE V. The expectation values of the dipole moment function, $\langle\mu\rangle$, in atomic units (1 D = 2.5418 a.u.) for vibrational levels v (all with $J=0$) in the $X\,^1\Sigma^+$ state of ^{7}LiH.

	$\langle\mu\rangle$		
v	Theory (this work	Exptl. (Ref. 11)	Exptl. (Ref. 12)
0	2.3380	2.314 ± 0.001	2.3141 ± 0.0001
1	2.3811	2.357 ± 0.001	2.3568 ± 0.0001
2	2.4226	2.399 ± 0.001	
3	2.4663		
4	2.5104		
5	2.5518		
6	2.5928		
7	2.6311		
8	2.6651		
9	2.6933		
10	2.7145		
11	2.7269		
12	2.7283		
13	2.7156		
14	2.6852		
15	2.6320		
16	2.5489		
17	2.4269		
18	2.2550		
19	2.0183		
20	1.6942		
21	1.2645		
22	0.7063		
23	0.0448		

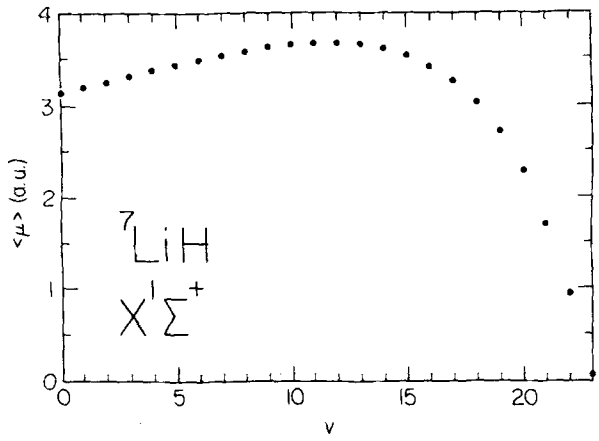


FIG. 6. The expectation value of the dipole moment, $\langle\mu\rangle$, in atomic units, for levels $v=0-23$ (all $J=0$) of the $X\,^1\Sigma^+$ ground state of ^{7}LiH. Note the rapid drop in $\langle\mu\rangle$ as dissociation is approached.

gence of the $v=23$, $J=0$ level. Lower v levels converge at smaller R_{max} values. However, even for converged calculations, it should be recognized that the dipole moment is also decaying rapidly with increasing distance and that this decay is fairly uncertain because of the sparcity of points in Table II at large distance.

The total Einstein $A_{v',J'}$ coefficient arising from all bound–bound transitions of the $X^1\Sigma^+ - X^1\Sigma^+$ band sys-

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TABLE VI. The strongest absorption (largest $|\mu|^2$ value) and strongest emission (largest A value) among the vibrotational levels of the $X\,^1\Sigma^+$ state of ^{7}LiH often do not correspond to the harmonic oscillator selection rule ($\Delta v = \pm 1$) but rather to higher $|\Delta v|$. Results here are all for a $J=0$ initial state (hence $J=1$ final state) but similar results are found for other J.

	Largest $\|\mu\|^2$		Largest A coefficient				
v ($J=0$)	$\Delta v=1$	$\Delta v=2$	$\Delta v=-1$	$\Delta v=-2$	$\Delta v=-3$	$\Delta v=-4$	$\Delta v=-5$
0	×		×				
1	×		×				
2	×		×				
3	×		×				
4	×		×				
5	×		×				
6	×		×				
7	×		×				
8	×		×				
9	×		×				
10	×		×				
11	×			×			
12	×			×			
13		×		×			
14		×		×			
15		×		×			
16		×		×			
17	×			×			
18	×			×			
19	×				×		
20	×				×		
21	×	...			×		
22	...	...				×	
23	...	...					×

tem may be obtained simply from the expression

$$A_{v',J'} = \frac{1}{2J'+1}\left((J')\sum_{v''=0}^{v'} A_{v',J';v'',J'-1} + (J'+1)\sum_{v''=0}^{v''=1} A_{v',J';v'',J'+1}\right)$$

(except that only the second term arises for $J'=0$). The reciprocals of these $A_{v',J'}$ values are the radiative lifetimes, $\tau_{v',J'}$, which are given in Table IV for $J'=0$, 1, and 10 for all bound v'.

The spontaneous radiative lifetimes in Table IV are of considerable interest. The dramatic increase in τ in the long-range level $v=23$, $J=0$ occurs as expected.[3] The effect of rotation is seen to be fairly small in most cases. For $v'=0$, the pure rotation emission rate increases by over three orders of magnitude in going from $J'=1$ to $J'=10$. The main reason for the lowering of $\tau_{v',10}$ compared to $\tau_{v',0}$ and $\tau_{v',1}$ for $v'=1, 2$, and 3 is the significance of the contribution of the pure rotational transition to the overall lifetime. For higher v', the lifetime increases monotonically as J' increases, with a dramatic increase in $v'=21$. There appear to be no other calculations or measurements of these lifetimes.

The dipole moment expectation values in Table V and Fig. 6 show an initial slow increase with v [as $\langle R\rangle$ increases and $\mu(\langle R\rangle)$ increases]. However, as dissociation is approached, μ must drop to zero and this is shown in particular by the order of magnitude decrease in going from $v=22$ to $v=23$. The experimental values[11,12] for $v=0$, 1, and 2 agree fairly well, but not precisely with theoretical values (as previously noted[5]) whether experimental potential curves are used, as here, or theoretical ones[5] (where $\langle\mu\rangle=2.350$, 2.559, and 2.711 a.u. were calculated for $v=0$, 5, and 10, respectively, fairly close to the present values in Table V).

V. DISCUSSION

One of the goals of our LiH work is to find out how realistic diatomic molecules (not harmonic or Morse oscillators) with realistic dipole moments (not constant or linear moments) behave. We feel this $X^1\Sigma^+$ LiH study does just that. In particular, with regard to radiative transition probabilities, we may contrast the harmonic oscillator selection rule, $\Delta v=\pm 1$, with what is observed in Table III and Figs. 3 and 4. We summarize this in Table VI, where it is shown that roughly the top half of the levels in the potential well do *not* even have the fundamental as their strongest transition in emission or absorption, let alone the only transition.

Unfortunately, there is apparently only one study[10] of the vibrational absorption spectrum in the $X^1\Sigma^+$ ground state of ^{7}LiH, and no emission studies. The uncertainty in the linewidth and other uncertainties make it difficult to convert the "line strength" reported[10] to the relative absorption coefficients implied by this work. However, we note that the dipole derivative estimated from experimental relative absorption intensities[10] "agrees exceptionally well" with that calculated by Docken and Hinze,[5] and hence with our $\mu(R)$ function.

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